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Assignment :

1] Logistic Regression

It is a supervised learning algorithm used for classification problems where the output is categorical. It uses a sigmoid functions to map predicted values into probabilities

* When to use :
  → Binary classification
  → Probability-based predictions.
  → Medical diagnosis, fraud detection.

* Method in Mathematics :

$$P(y=1) = \frac{1}{1+e^{-z_s}}$$

where $z_s = b_0 + b_1 x$.

* Working :
  i] Initialize weights
  ii] Compute linear equation $z_s = b_0 + b_1 x$.
  iii] Apply sigmoid function
  iv] Compute error using loss function
  v] Update weights using gradient descent
  vi] Repeat untill convergence.

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Example:

Dataset:

x	1	2	3
y	0	0	1

Assume: $b_0 = -2$, $b_1 = 1$.

For $x = 3$:

$$z = -2 + (1 \times 3) = 1$$

$$P = \frac{1}{1+e^{-1}} = 0.731$$

since $P > 0.5$, output $= 1$.

3 Support vector Machine (SVM):

SVM is a supervised learning algorithm used for classification & regression that finds the optimal hyperplane seperating classes.

* When to Use:
  - → High dimentional data
  - → Image classification
  - → Text classification

* Mathematical Model:

  Hyperplane: $wx + b = 0$.

* Working:
  - → Plot data points
  - → Identify support vectors
  - → Find hyperplane maximizing margin.
  - → Classification new points based on position.

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Example:

Dataset:

x	2	4	1
y	3	5	1
class	+1	+1	-1

Hyperplane separates classes → decision boundary computed.

3] K- Nearest Neighbors (KNN):

It is a supervised learning algorithm that classifies data based on similarity.

* When to use:
  - → Small dataset
  - → Pattern recognition
  - → Recommendation systems.

* Formula:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

* Working steps:
  - i] choose K.
  - ii] Calculate distance
  - iii] Selecting nearest neighbors.
  - iv] Perform majority voting.

Example:

Dataset:

x	1	2	5
y	1	2	5
class	A	A	B

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Testpoint : (3,3), K = 3.

Distances :

→ $d_1 = \sqrt{(3-1)^2 - (3-1)^2} = 2.82$

$d_2 = 1.41$

$d_3 = 2.82$

Nearest : A, A, B → Majority = A.

4) Navie Bayes Algorithm :

It a probabilistic classifer based on Bayes theorem assuming feature independente.

* ~~Bayes to~~ Formula :

$P(A|B) = P(B|A) \times P(A) / P(B)$.

* Working Steps :

→ Calculate prior probabilities.

→ Compute likehood

→ Apply Bayes theroem.

→ Select class with highest probability.

Example :

Data set :

class	Yes	NO
count	6	4

$P(yes) = 0.6$, $P(NO) = 0.4$

$P(X|yes) = 0.5$, $P(X|NO) = 0.2$.

$P(yes|X) = 0.5 \times 0.6 = 0.3$.

$P(NO|X) = 0.2 \times 0.4 = 0.08$.

Prediction = Yes.

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5] Decision Tree:

It is a supervised learning algorithm that splits into branches to make decision.

* When to use:
  → Interpretability required.
  → Classification & regression problems.

* Formula:

$$\text{Entropy} = -p \log_2 p - q \log_2 q$$

* Working steps:
  → Select best feature
  → Split dataset
  → Repeat recursively
  → Stop at leaf node.

Example:

Dataset : 10 samples → Yes 6 , NO 4.

$$\text{Entropy} = 0.6 \log_2 0.6 + 0.4 \log_2 0.4 = 0.97.$$

6] Random Forest:

It is an ensemble learning method that combines multiple decision trees for better accuracy.

* When to use:
  → Avoid overfitting
  → Large dataset
  → High accuracy needed.

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* Formula :

$$g(x) = f_1(x) + f_2(x) + \dots + f_n(x).$$

* Working steps :

i] Create multiple dataset
ii] Train decision trees
iii] Aggregate results
iv] Output final prediction

Example :

3 predictions :

→ Tree 1 → Yes
→ Tree 2 → Yes
→ Tree 3 → No

Final output = Yes (majority voting)

7] Confusition Matrix :

	Predction (+)	Predction (−)
Actual (+)	TP	FN
Actual (−)	FP	TN

* Formulas :

→ $\text{Accuracy} = \dfrac{TP + TN}{\text{Total}}$

→ $\text{Precision} = \dfrac{TP}{TP - FP}$

→ $\text{Recall} = \dfrac{TP}{TP + FN}$

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→ F1 Score = $\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Example:

Given, TP = 40
TN = 50
FP = 5
FN = 5.

Accuracy = 40+50/100 = 0.9

Precission = 40/45 = 0.88

Recall = 40/45 = 0.88.

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